Team-Members
Devanshi Shah
Arietta Goshtasby

Introduction

Team-Members
Ronjan Husain Mian
Ariadne Colina-Valeri

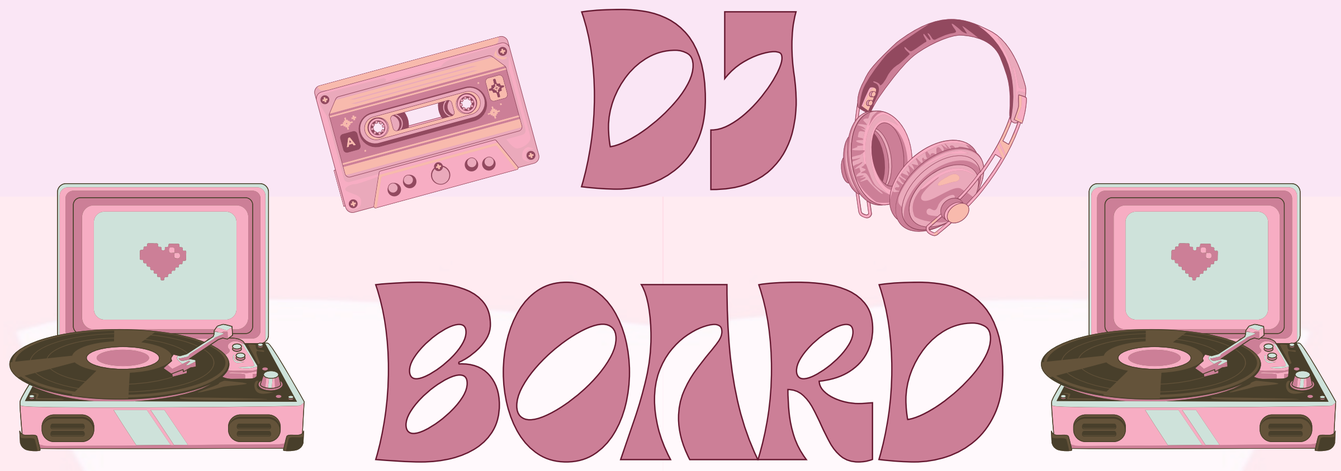
This project is a circuit made audio effects processor built on the STM32L432KC microcontroller. Press a button, speak or play sound into the microphone, and the device records, processes, and plays it back through a speaker with your chosen effects like echo or reverse. It is a low cost system that handles real-time digital signal processing. Record your own audio and generate unique beats on the spot!

Effects Made

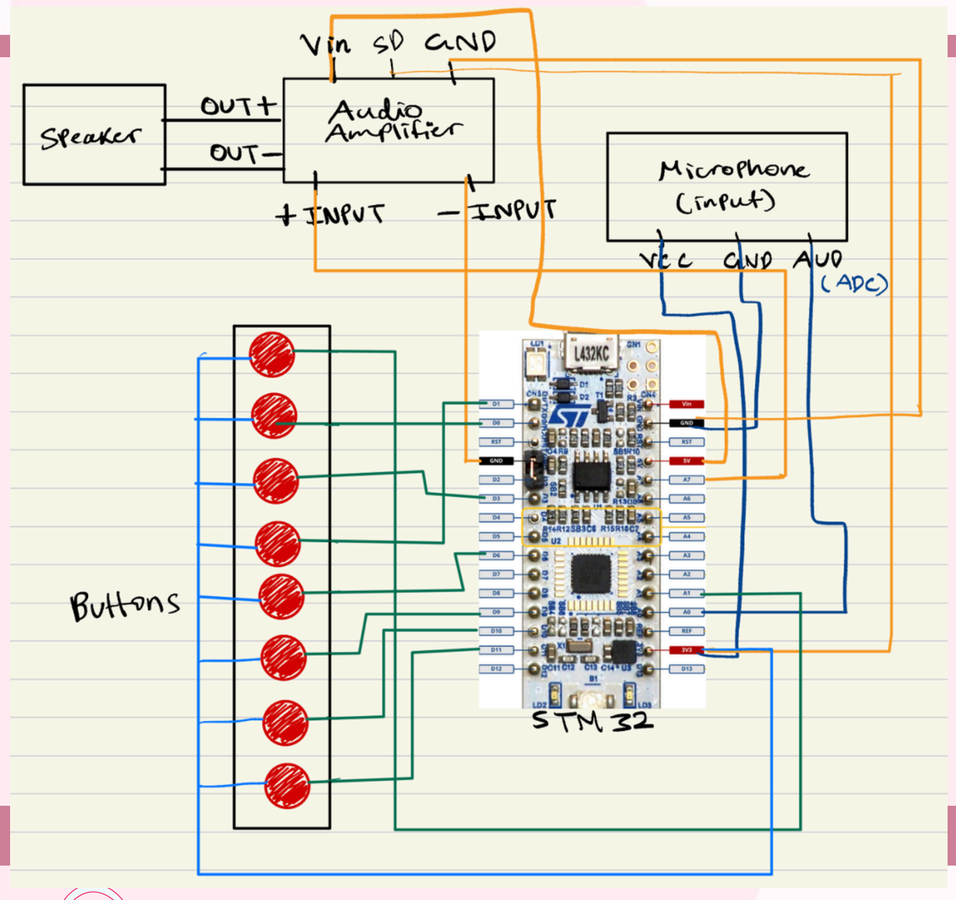
1.) Echo
2.) Reverse
3.) Speed up
4.) Loop-Size
5.) Pitch-Up
6.) Pitch-Down
7.) Volume-Up

Signal Chain

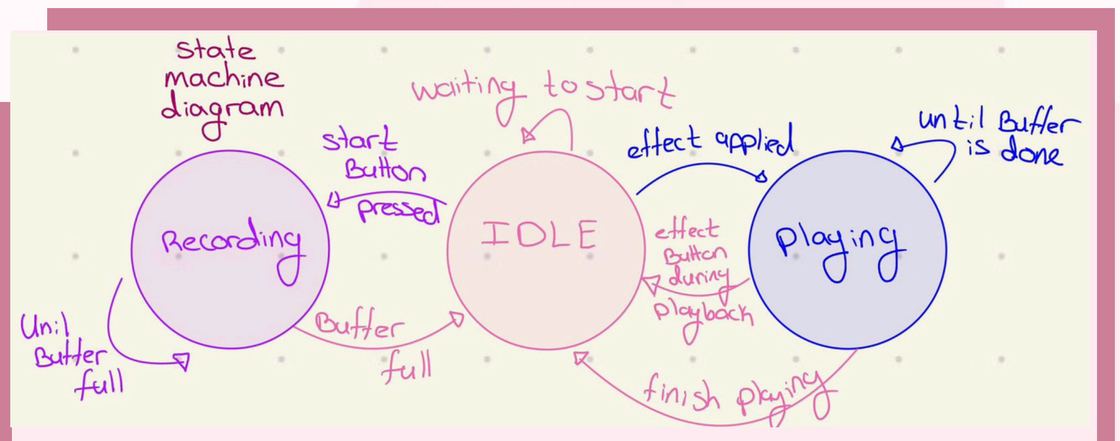
MIC → ADC → STM32 buffer → effects → DAC → Audio Amp → Speaker



Hardware

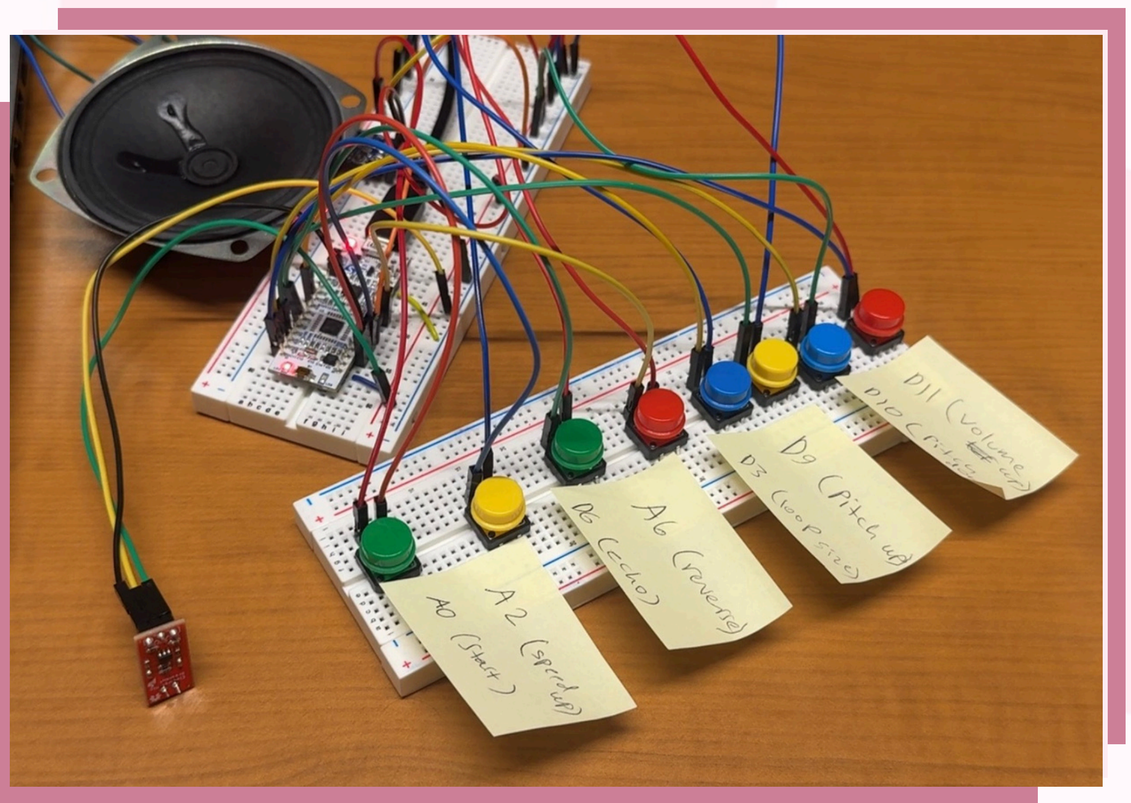


Software

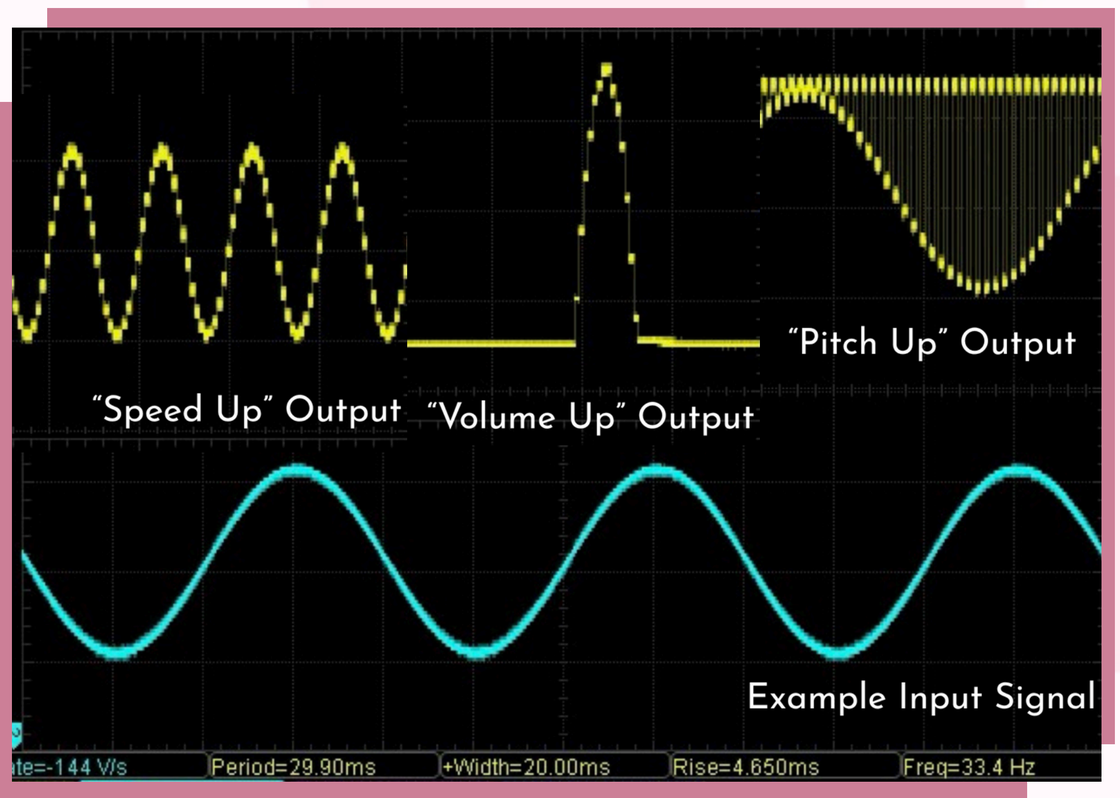


All effects (echo,reverse,pitch,etc.) are applied directly to the buffer before playback.

Circuit



Visual



Tufts UNIVERSITY | School of Engineering
Electrical and Computer Engineering